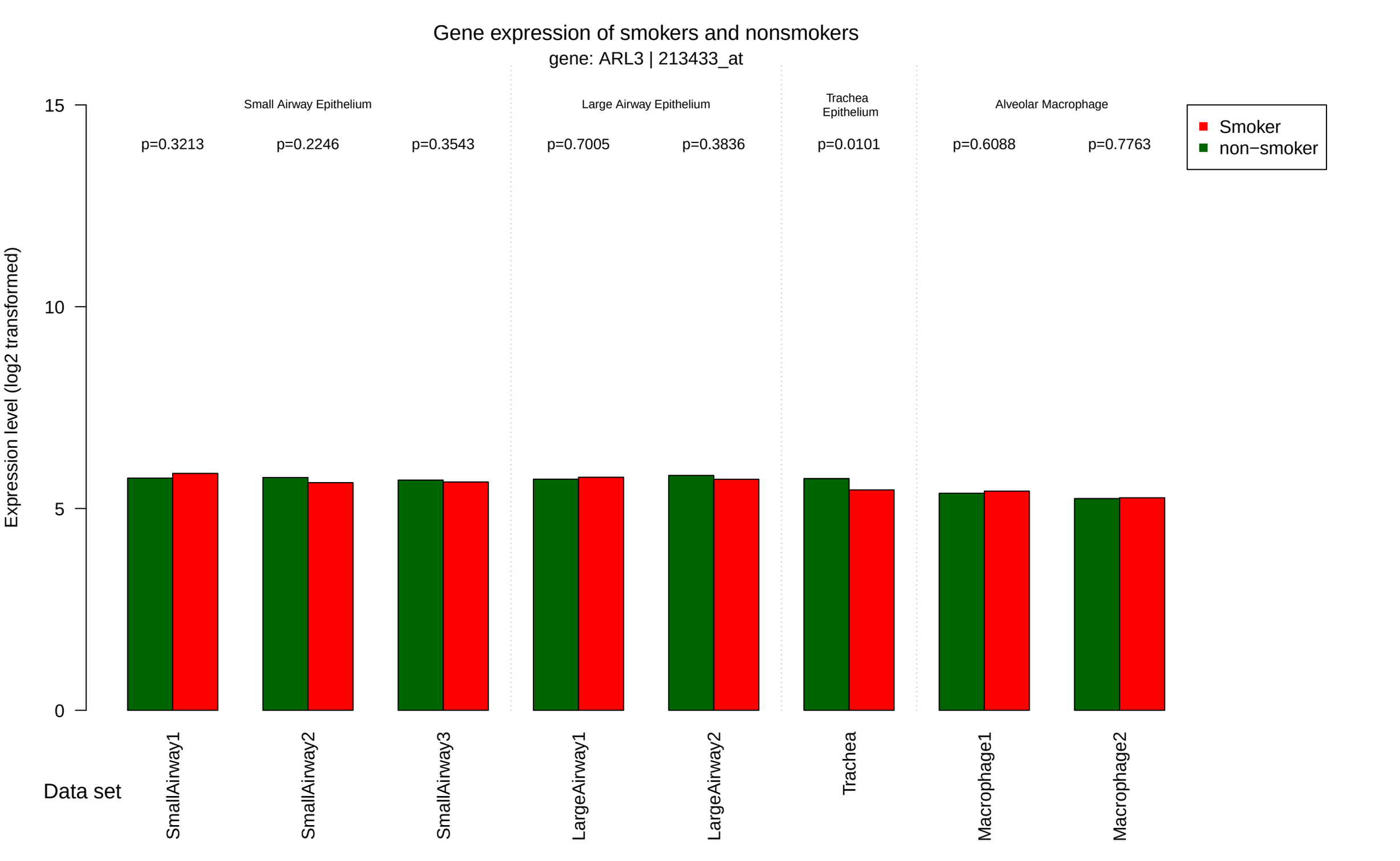
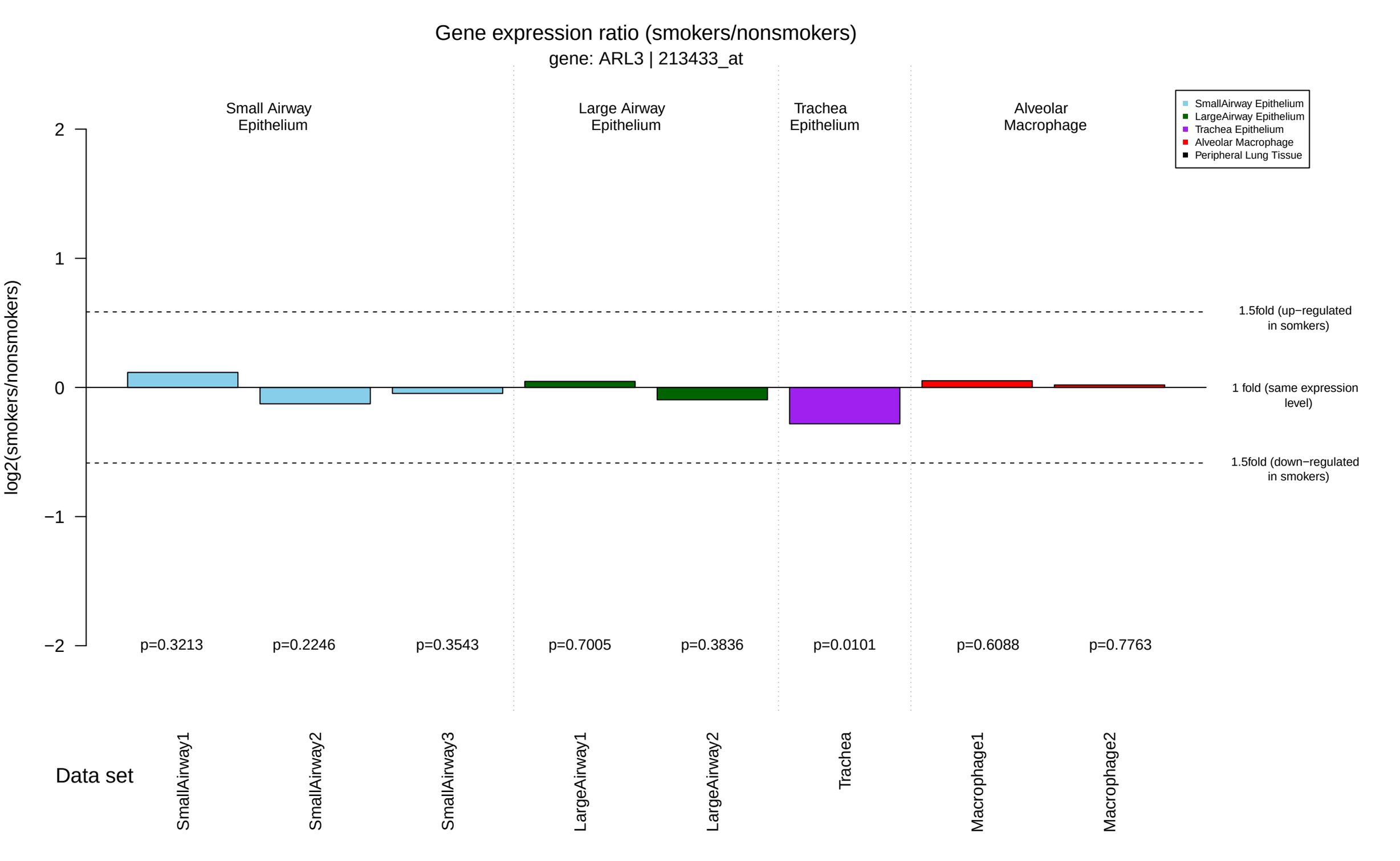
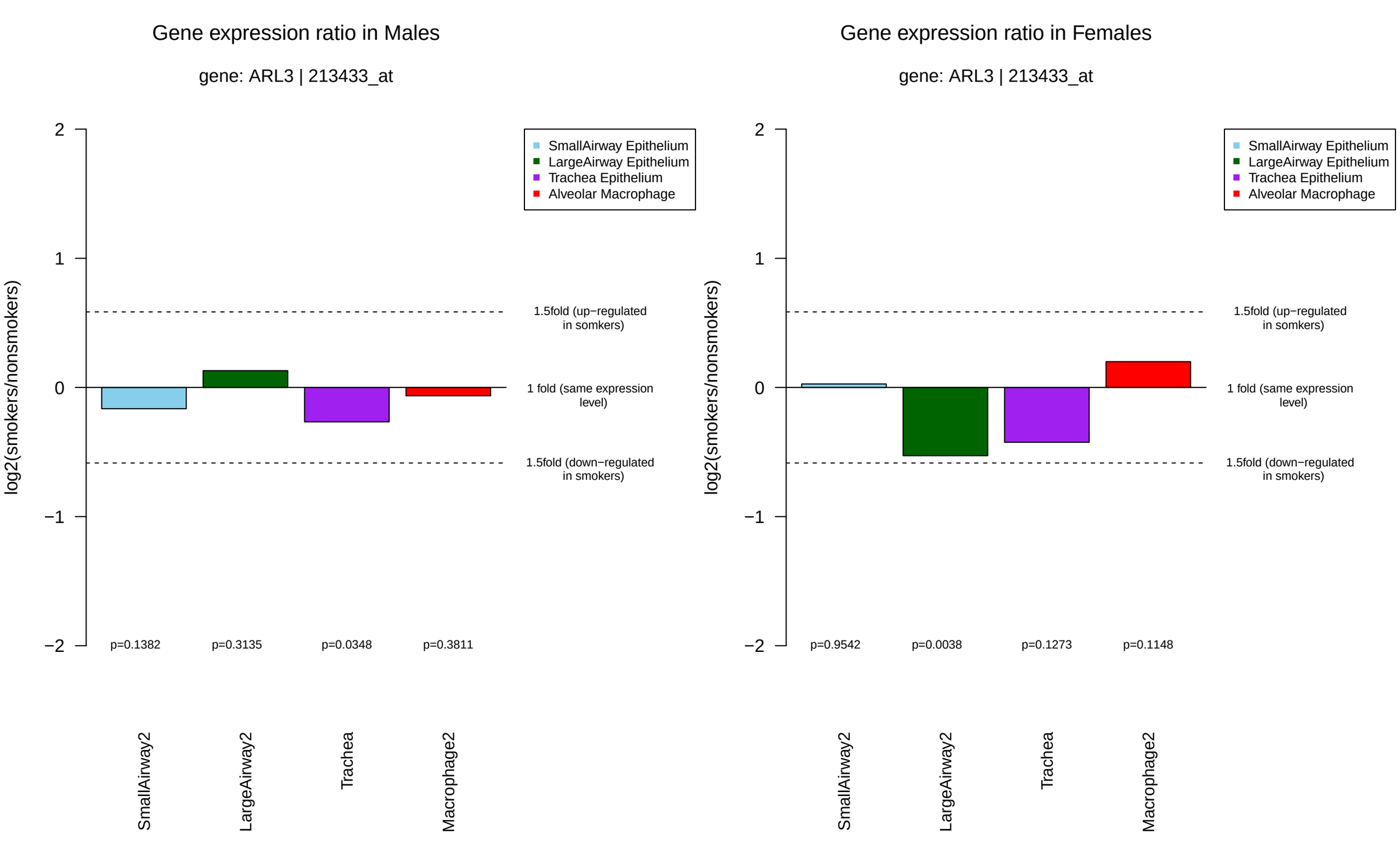
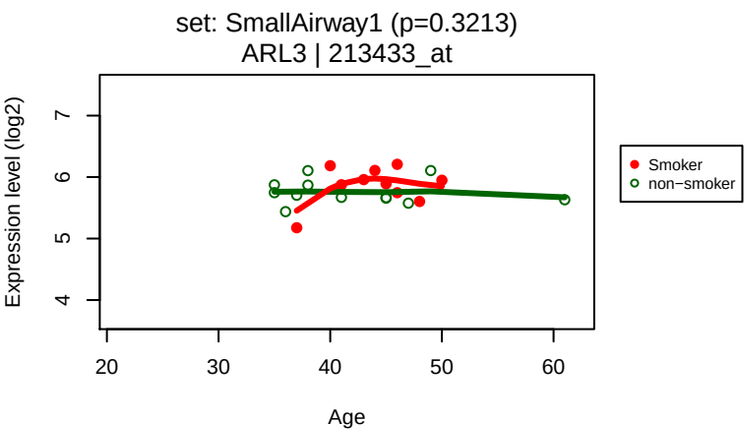
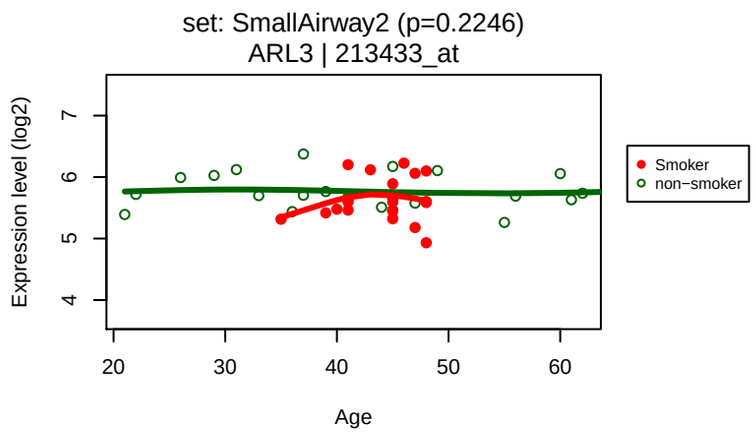
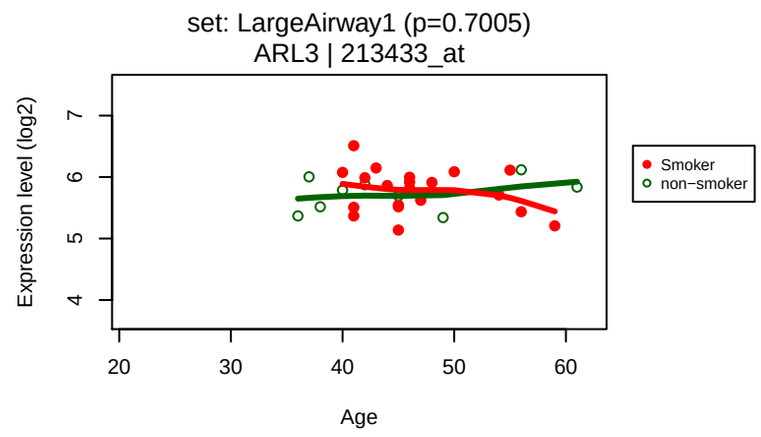

Gene expression of smokers and nonsmokers
gene: ARL3 | 213433_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3213
p=0.2246
p=0.3543
p=0.7005
p=0.3836
p=0.0101
p=0.6088
p=0.7763
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

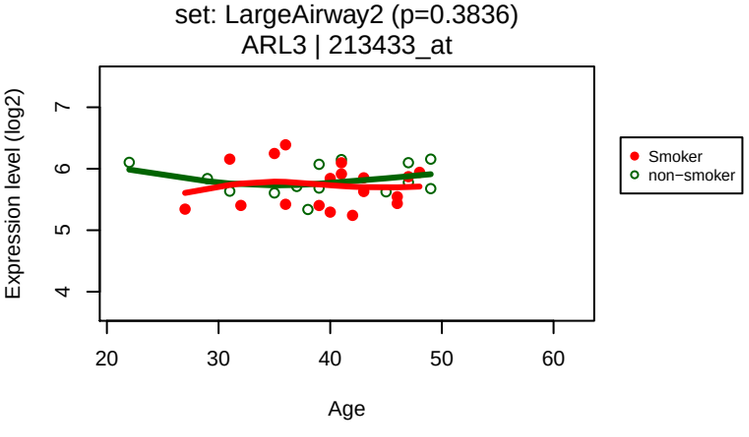

Gene expression ratio (smokers/nonsmokers)
gene: ARL3 | 213433_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3213
p=0.2246
p=0.3543
p=0.7005
p=0.3836
p=0.0101
p=0.6088
p=0.7763
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

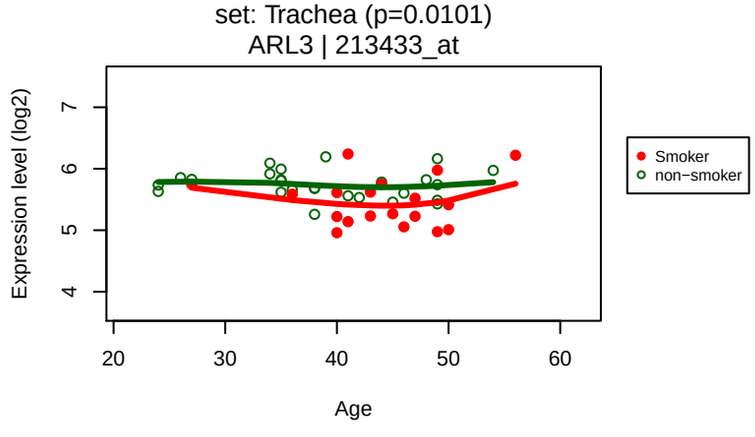

Gene expression ratio in Males
gene: ARL3 | 213433_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1382
p=0.3135
p=0.0348
p=0.3811
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL3 | 213433_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9542
p=0.0038
p=0.1273
p=0.1148
SmallAirway2
LargeAirway2
Trachea
Macrophage2

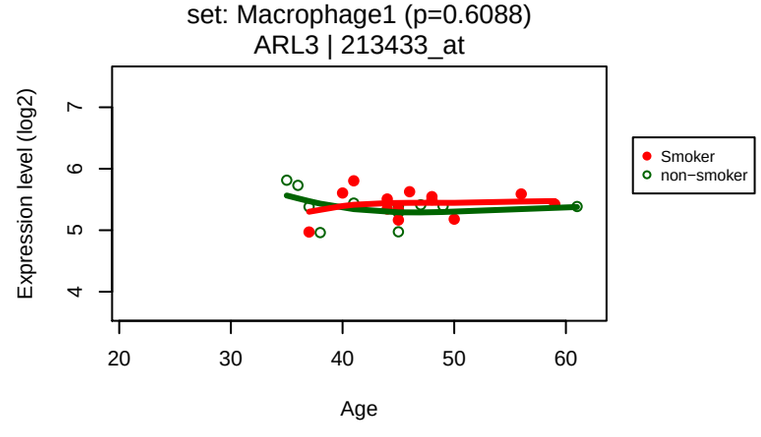

set: SmallAirway1 (p=0.3213)
ARL3 | 213433_at
Expression level (log2)
Age
Smoker
non−smoker

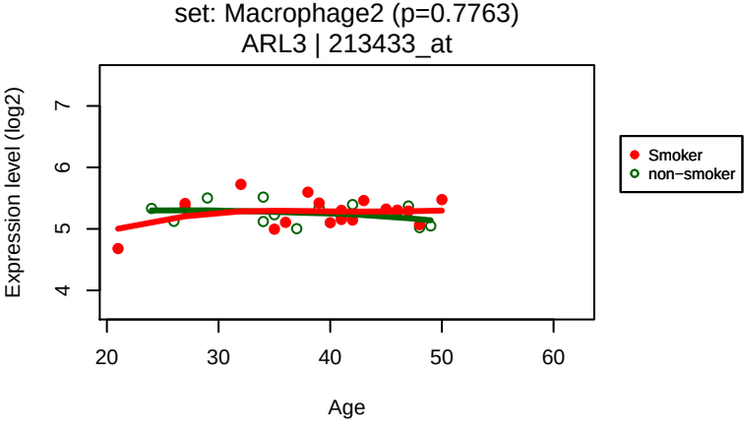

set: SmallAirway2 (p=0.2246)
ARL3 | 213433_at
Expression level (log2)
Age
Smoker
non−smoker

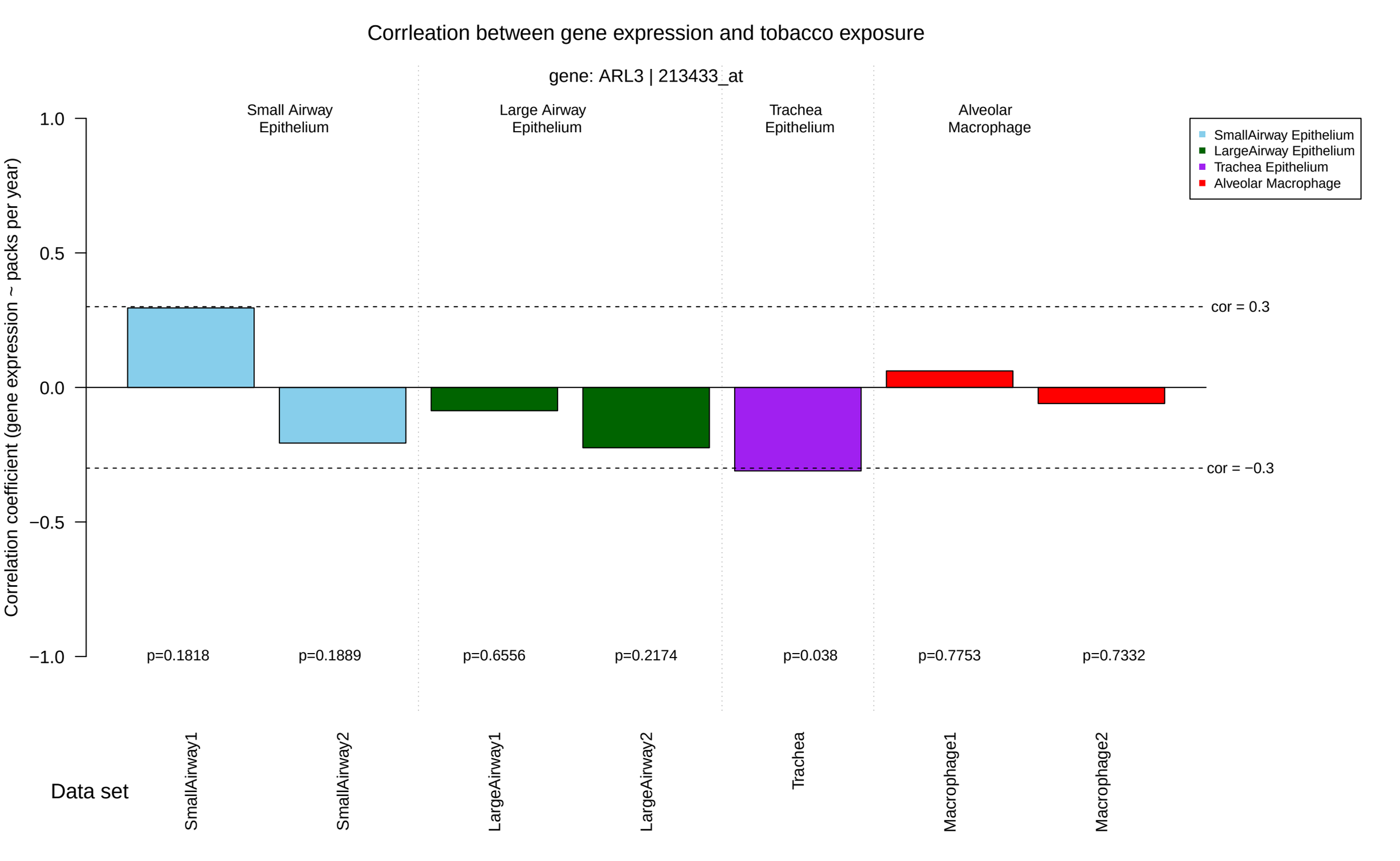

set: LargeAirway1 (p=0.7005)
ARL3 | 213433_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3836)
ARL3 | 213433_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0101)
ARL3 | 213433_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6088)
ARL3 | 213433_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7763)
ARL3 | 213433_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL3 | 213433_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1818
p=0.1889
p=0.6556
p=0.2174
p=0.038
p=0.7753
p=0.7332
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2